Dismantling Workplace Norms: Strategies for Cultural Diffusion

Alastair Langtry* DJ Thornton†

January 15, 2025

EXTENDED ABSTRACT
Submission invited by Pauline Grosjean

1 Introduction

Organizational culture and norms inside firms have a large impact on their performance. They affect cooperation between employees, generation and scrutiny of new ideas, and compliance with regulations. Ultimately, better culture can make a firm more profitable— as well as creating a happier and healthier environment for its stakeholders. Given this, it is not surprising that firms devote significant time and effort to shaping their culture and promoting behavioral norms among their employees.

We consider the problem faced by an organization who wants to replace an existing norm with a new one. For example, the organization might want to encourage a new norm of inclusivity in place of some norm of discriminatory behavior (e.g. jokes about a minority group; bullying). Or the organization could have a norm in which some regulation is not taken seriously by employees (e.g. coworkers share passwords with each other), and the organization would prefer that employees take the regulation seriously.

In our model, agents (employees) interact in an organizational network and face local complementarities to adoption of the old norm. A key feature of our model is that the organizational network is multiplexed, meaning that agents can have different types of links that serve distinct purposes. We distinguish between links in the "formal network" which drive productivity by enabling collaboration and knowledge-sharing, and links in the "informal network" which shape interpersonal behaviors and enforce social norms. This multiplexed structure creates a tradeoff: cutting links to disrupt the old norm may inadvertently weaken productivity. Hence the organization's op-

*University of Bristol, UK (email: `alastair.langtry@bristol.ac.uk`). This work was supported by the Economic and Social Research Council [award reference ES/P000738/1]. Any remaining errors are the sole responsibility of the authors.

†University of New South Wales, Australia (email: `d.thornton@unsw.edu.au`).

timal policy must account for the interaction between the formal network and the informal network among employees.

We assume that agents prefer to maintain the old norm when at least k of their neighbors in the informal network do the same. All agents are made aware of a new norm at time 0 and update their best responses myopically, observing in each period the actions of their neighbors. In the unique Nash equilibrium of the game, agents maintain the old norm if and only if they belong to the *k-core* of the informal network—the largest subgraph in which every agent has at least k neighbors.[1]

We begin by characterizing the extent of diffusion of the new norm as a function of primitives on the network structure—namely, features of the *degree distribution*—leveraging results from the theory of random graphs. In the limit of a large network, a simple condition—the existence of a *giant k-core* in the informal network— determines whether a nonzero fraction of agents adopt the old norm in the *limit-equilibrium*.

We then examine the effects of different interventions on reducing the prevalence of the old norm. For example, firms may utilize "seeding" technologies (e.g., inclusion seminars) to directly promote adoption of the new norm, or they may invest in monitoring technologies (e.g., computer surveillance) to disrupt the old norm. Importantly, differences in how these technologies interact with the multiplex nature of the network can lead to subtle and unexpected consequences in equilibrium. For example, if better monitoring tools reduce the persistence of the old norm, firms may respond by increasing the connectivity among their employees. In such cases, looking only at raw observational data might underestimate the effectiveness of interventions, as the endogenous response from the firm would obscure the reduction in links reinforcing the old norm. Our framework provides insights into these dynamics and the conditions under which multiplexed networks are more or less advantageous for the firm. This allows us to provide a number of policy recommendations for an organization attempting to reduce the prevalence of the old norm.

Our work contributes to the literature on organizational culture and the diffusion of behaviors in networks.

2 Model

For the purposes of this extended abstract, we focus only on the informal network and present our results on equilibrium behavior. We consider a sequence of games, $\{\Gamma^{(n)}\}_{n\in\mathbb{N}}$, and set out the game for a given n.

Agents & Endowments. There are n agents $i \in N = \{1, ..., n\}$. Before the game starts, each pair of agents independently form a link with probability $p = \lambda/n$, for some $\lambda \geq 0$. This means

[1] This is a known result for threshold games on finite graphs (e.g. see Gagnon and Goyal [2017], Langtry et al. [2024]).

that agents are embedded in an unweighted and undirected Erdös-Renyi random network, denoted $G(n,p)$. We say i and j are *friends* if $G_{ij} = 1$. Additionally, to begin all agents follow an *old norm*.

Actions & Preferences. At the start of the game ($t = 0$), a *new norm* arrives and all agents learn that it exists.[2] So each i has two possible actions: maintain the old norm ($a_{i,t} = 1$) or adopt the new norm ($a_{i,t} = 0$). Actions are *strategic complements*: following a given norm is more attractive to agent i when more of her friends follow it. Specifically, we assume that agents prefer to follow the old norm if and only if at least k friends follow it too. Agents' preferences can be represented by a utility function $u(a_i, M_i)$ such that:

$$u\left(1, M_i\right) \geq u\left(0, M_i\right) \iff M_i \geq k, \quad \text{where} \quad M_i = \sum_j G_{ij} a_j. \tag{1}$$

We assume agents break indifferences in favour of the new norm.

Timing, Dynamics & Information. The game $\Gamma^{(n)}$ is played over $n+1$ time periods. At period $t = 0$, all agents are following the old norm: $a_{i,0} = 1$ for all i. Nature realizes the network $G = G(n,p)$ and agents learn of the new norm. At each subsequent period $t \geq 1$, agents myopically best respond to behaviour from the previous period. This yields best responses:

$$BR_{i,t}^{(n)}(M_{i,t}) = \begin{cases} 1 & \text{if } M_{i,t-1} \geq k \\ 0 & \text{if } M_{i,t-1} < k \end{cases} \tag{2}$$

Agents know of the new norm's existence, and observe the histories of their friend's actions. Under myopic best responses, this information is sufficient for agents to choose optimally at every t.[3]

Equilibrium. An action profile $(a_1, ..., a_n)$ is a Nash Equilibrium of the game $\Gamma^{(n)}$ if and only if

$$(a_1, \ldots, a_n) = \left(BR_1^{(n)}(M_1), \ldots, BR_n^{(n)}(M_n)\right). \tag{3}$$

Discussion. We think of the *old norm* as a behavior or attitude that has been held for a long time. Examples might include exclusionary practices or inefficient processes. The *new norm* represents an alternative behavior that is costly to undertake when surrounded by others who hold to the old norm, but is preferred when enough others abandon the old norm, for example jokes about a

[2]This is not an important assumption. We could equally assume that agents learn of its existence with some positive probability, and then have this information diffuse through the network. This would complicate the analysis somewhat but would not affect the qualitative substance of any of our results.

[3]That is, giving agents additional information would also have no impact on behavior. This is because the actions of i's neighbors at time $t-1$ are necessary and sufficient for i to choose optimally at time t.

minority group only bring utility if they receive a sufficiently large audience. Our utility specification is flexible enough that the new norm can also be to simply abandon the existing norm for its "opposite", i.e. lack of distasteful jokes.

The way agents interact with each other matters for understanding how cultural behaviors spread, because these behaviors often depend on peer influences. Using a random graph to model the interaction structure affords us a degree of tractability in the analysis. In particular the Poisson random graph allows us to parameterize the connectivity of the network and therefore the extent to which local complementarities are available to agents. This in turn allows us to derive sharp comparative statics by using well-understood transformations of the degree distribution.

2.1 Threshold Games

The game $\Gamma^{(n)}$ is a *threshold game* – agents have binary actions, and take one of the actions if the number of their friends taking that action exceeds some threshold. In our setting, the threshold: (1) is an absolute number of friends, and (2) is common to all agents. In models with this feature, the notion of a *k-core* is crucial to the analysis. The k-core of a network G is the largest sub-network H of G such that every agent in H has at least k friends in H [Seidman, 1983, Frieze and Karoński, 2016].[4] In fact, in our model, it is all that we need to characterise the equilibrium.

Remark 1. *The game $\Gamma^{(n)}$ has a unique Nash Equilibrium. Agents follow the old norm if and only if they are in the k-core of the network G.*

We can see how the k-core characterises equilibrium behaviour directly from the best response dynamics. Write $\mathcal{M}_t \subseteq V(G)$ for set of agents who *maintain* the old norm at time t, i.e. $a_{it} = 1$ if and only if $i \in \mathcal{M}_t$. By assumption, all agents maintain the old norm at $t = 0$, so $\mathcal{M}_0 = V(G)$. At $t = 1$, agents move to the new norm if they have fewer than k friends. It is clear that these agents will stick with the new norm at all future periods. When thinking about who follows the old norm, we can discard these agents and focus on the induced subgraph among the remaining set of agents, $\mathcal{M}^1$.

The same thing happens in the next period. At $t = 2$, agents move to the new norm if they have fewer than k friends inside the set $\mathcal{M}^1$ As before, these agents will stick with the new norm at all future periods. So we can discard them from $\mathcal{M}^1$ to give us $\mathcal{M}^2$. This process iterates until $\mathcal{M}^{t+1} = \mathcal{M}^t$ (and it must come to an end by monotone convergence). At that point, all remaining agents have at least k friends inside the set of remaining agents (who are still maintaining the old norm). So they also maintain the old norm, and keep doing so. This is exactly the definition of the k-core.

Remark 1 is not in itself novel. Threshold games have already been studied in cases where the network is fixed, and the link between equilibrium behaviour and the k-core established [Gagnon

[4] In graph theory terms, H is k-connected.

and Goyal, 2017, Langtry et al., 2024]. But it shows that the question of who follows which norm can reduced to a question about the size/structure of the k-core. We are able to answer these questions precisely in the limit as $n \to \infty$.

The rest of our analysis will focus on the the limit as $n \to \infty$. As such, the equilibria we characterize are *limit-best-replies* as a function of the underlying network structure.[5] This allows us to present precise comparative statics, in contrast to the literature on finite threshold games in which comparative statics results are usually of the form "anything goes".[6]

3 Results

A natural starting point is to ask (1) whether the old norm will persist, or whether it will be overrun by the new norm, and (2) if the old norm does persist, what fraction of agents will stick with it?[7]

Answering these questions amounts to understanding whether the network contains a *giant* k-core, and how big it is. A giant k-core is a k-core containing a positive *fraction* of all agents. It turns out that the existence of a giant k-core exhibits a sharp threshold with a markedly simple characterization,[8] and its size is the solution to a relatively simple implicit function. This characterization allows us to talk about the properties of equilibrium as a function of the density of the network as captured by λ. We say that the old norm is *persistent* if a positive fraction of agents maintain the old norm (with high probability) in the unique limit equilibrium. We refer to the fraction of agents adopting a norm as that norm's *prevalence*.

Theorem 1. *Let* $\psi_k(\lambda) = \mathbb{P}(\mathrm{Po}(\lambda) \geq k)$ *and define* $c_k \equiv \min_{\mu \geq 0} \frac{\mu}{\psi_{k-1}(\mu)}$. *Then,*

(i) The old norm is persistent if and only if $\lambda > c_k$.

(ii) The prevalence of the old norm is given by $\psi_k(\lambda\hat{\rho})$, *where* $\hat{\rho}$ *is the largest solution in* $[0, 1]$ *to the equation* $\hat{\rho} = \psi_{k-1}(\lambda\hat{\rho})$.

Theorem 1 is a direct application of Janson and Luczak, 2007, Thm 2.3 and Thm 2.4.[9] Our focus is on the comparative statics of equilibrium persistence and prevalence, to which we now turn. If the old norm is not persistent, then by definition its prevalence is 0. Moreover, because the

[5] Note that the limit best-replies are unique by remark 1 so the equilibrium is well defined.

[6] See for example, Gagnon and Goyal [2017, Theorem 2], and [Langtry et al., 2024, Remark 2(i)] – both of which are special cases of Milgrom and Shannon [1994, Theorem 13].

[7] Clearly if there is a nontrivial fraction of agents adopting the new norm, then the number of agents who adopt diverges to infinity which is precisely why the number of adopting agents is of less interest than the fraction of adopting agents in our setting.

[8] A sharp threshold for a graph property P in $G(n, p(n))$ is function $p^*(n)$ such that for all $\epsilon > 0$, if $p(n)/p^*(n) < 1-\epsilon$ then the graph has the property P with probability $o(1)$, while if $p(n)/p^*(n) > 1+\epsilon$ then the graph has the property with probability $1 - o(1)$. It is well known that every monotone graph property has a sharp threshold, and the presence of a k-core is a monotone graph property.

[9] The result for Erdös-Renyi random graphs graphs goes back to Pittel et al. [1996], but Janson and Luczak [2007] provide a particularly clean characterization of a more general result.

k-core exhibits a sharp threshold, a slight increase in the density (λ) of the network will not move the prevalence away from 0. Hence the equilibrium is essentially[10] unresponsive to changes in λ when λ is small. In a persistent equilibrium on the other hand, we have the following proposition.

Proposition 1. *In any persistent equilibrium,*

(i) The prevalence of the old norm is strictly increasing in λ, and

(ii) The prevalence of the old norm is strictly decreasing in the adoption threshold k.

Proposition 1 is very intuitive: part *(i)* says that as the network gets more dense, it becomes more difficult for agents to abandon the old norm since they have (on average) more friends who adopt it in each time period. Part *(ii)* says that the stronger the complementarity required to maintain a norm, the smaller the prevalence of that norm will be.

4 Organizational Policy and Targeting

We now turn to the question of how the organization can maximize the prevalence of the new norm. For the purposes of this extended abstract we restrict our attention to the effects of *seeding* the new norm, and as mentioned before, we will not consider here the tradeoff induced by multiplexing (i.e. whether the firm would actually *want* to engage in seeding). We first introduce a notion of equilibrium robustness which allows us to talk precisely about the effect of seeding. In what follows we assume that the old norm is persistent in equilibrium (i.e. in the Poisson model, that $\lambda > c_k$).

4.1 Robustness and Seeding

We now formalize the idea of equilibrium robustness. Suppose the firm can seed a fraction $\sigma \in [0, 1]$ of agents with the new norm.[11] One can think of this as though the firm tries to convince all agents to accept the new norm and is successful with probability σ independently for each agent. In our analysis it doesn't matter for seeding whether the firm acts at the beginning of the game or after equilibrium has been reached, so for simplicity we suppose the firm acts at $t = 0$. This kind of seeding is equivalent to *site percolation* of the random graph where vertices are "occupied" with probability σ.

Definition 1. *The old norm is σ-robust if it is persistent in equilibrium w.h.p. to σ-seeding by the firm.*

The notion of σ-robustness captures how difficult it is for the firm to disrupt a persistent equilibrium via random seeding. It turns out that robustness of equilibrium has a markedly simple characterization, which we now present.

[10] Aside from changes among a finite number of agents.

[11] $\sigma = 0$ encompasses finite seeding.

Theorem 2. *Equilibrium is $\hat{\sigma}$-robust if and only if the expected degree of a randomly chosen agent after $\hat{\sigma}$-seeding is at least c_k, that is if and only if $\hat{\sigma} < 1 - c_k/\lambda$.*

Put differently, Theorem 2 says that a firm who wants to quash the old norm *must* seed at least $1 - c_k/\lambda$ agents with the new norm. We refer to $\sup_{\sigma>0}\{\text{equilibrium is } \sigma\text{-robust}\}$ as the minimal seeding capacity required to disrupt a persistent norm. Then Theorem 1 implies the following.

Corollary 1. *The minimal seeding capacity required to disrupt a persistent norm is strictly increasing in network density λ and strictly decreasing in the threshold k.*

Corollary 1 complements existing results on seeding a simple contagion (e.g. Akbarpour et al. [2020], Sadler [2025]). For a simple contagion, a successful seed is one which lands in the giant component (i.e. a giant 2-core). In our setting, a nontrivial fraction of agents must be *removed* from the giant k-core in order for equilibrium prevalence of the old norm to decline. This suggests that targeted seeding may substantially more effective than random seeding. Indeed, in our model every seed outside of the k-core is "wasted". In the context of organizations, one could think of random seeding as holding a large seminar in which employees are provided information about the new norm. Some agents may change their behavior, but it would be far more effective to target only those with the undesired behavior. Moreover, the organization doesn't need to target all agents with undersirable behavior, it only needs to target enough to dismantle (or at the very least reduce the size of) the k-core. Of course in reality this may not be feasible for legal or other reasons (e.g. targeting the actions of a particular group could appear discriminatory). The effectiveness of targeted seeding over random seeding turns out to depend on the exact degree distribution in the network.

References

M. Akbarpour, S. Malladi, and A. Saberi. Just a few seeds more: value of network information for diffusion. *Available at SSRN 3062830*, 2020.

A. Campbell, D. Thornton, and Y. Zenou. Strategic diffusion: Public goods vs. public bads. Technical Report DP19443, CEPR, 2024.

A. Frieze and M. Karoński. *Introduction to random graphs*. Cambridge University Press, 2016.

J. Gagnon and S. Goyal. Networks, markets, and inequality. *American Economic Review*, 107(1): 1–30, 2017.

S. Janson and M. J. Luczak. A simple solution to the k-core problem. *Random Structures & Algorithms*, 30(1-2):50–62, 2007.

A. Langtry, S. Taylor, and Y. Zhang. Network threshold games. Technical report, arXiv. org, 2024.

P. Milgrom and C. Shannon. Monotone comparative statics. *Econometrica: Journal of the Econometric Society*, pages 157–180, 1994.

B. Pittel, J. Spencer, and N. Wormald. Sudden Emergence of a Giant$k$-Core in a Random Graph. *Journal of Combinatorial Theory, Series B*, 67(1):111–151, May 1996. ISSN 0095-8956. doi: 10.1006/jctb.1996.0036. URL `https://www.sciencedirect.com/science/article/pii/S0095895696900362`.

E. Sadler. Seeding a simple contagion. *Forthcoming at Econometrica: Journal of the Econometric Society*, 2025.

S. B. Seidman. Network structure and minimum degree. *Social networks*, 5(3):269–287, 1983.

A Appendix

A.1 Proof of Proposition 1

We begin by showing the result with respect to λ. For any fixed $\lambda > \min_{\mu>0} \mu/\psi_{k-1}(\mu)$ and any $k \geq 3$,

$$\psi_k'(\lambda) = \frac{\lambda^{k-1}e^{-\lambda}}{(k-1)!}.$$

Hence for the implicit function $\lambda\hat{\rho} = \lambda\hat{\rho}(\lambda)$, we have by the chain rule that

$$\frac{d}{d\lambda}\psi_k(\lambda\hat{\rho}) = \left(\hat{\rho} + \lambda\hat{\rho}'\right)\psi_k'(\lambda\hat{\rho}).$$

Since $\psi_k'(\lambda) > 0$, the size of the giant k-core is increasing if and only if $\hat{\rho} + \lambda\hat{\rho}' > 0$. By implicitly differentiating, we can show that

$$\hat{\rho}' = \frac{\hat{\rho}\psi_{k-1}'(\lambda\hat{\rho})}{1 - \lambda\psi_{k-1}'(\lambda\hat{\rho})},$$

from which is it follows that the size of the giant k-core is increasing in λ iff

$$\hat{\rho}\left[1 + \frac{\lambda\psi_{k-1}'(\lambda\hat{\rho})}{1 - \lambda\psi_{k-1}'(\lambda\hat{\rho})}\right] \iff \lambda\psi_{k-1}'(\lambda\hat{\rho}) < 1.$$

To prove that the final inequality holds, we invoke Janson and Luczak, 2007, Lemma 7.1, 7.2. By Lemma 7.1. there exists two distinct positive solutions $\mu^-(\lambda) < \mu^+(\lambda) = \lambda\hat{\rho}$ to the equation $\mu/\psi_{k-1}(\mu) = \lambda$, and for any $\mu \in (\mu^-, \mu^+)$ we have $\mu/\psi_{k-1}(\mu) < \lambda$. From this it follows that $\lambda\hat{\rho} > \min_{\mu>0} \mu/\psi_{k-1}(\mu)$, and therefore by Lemma 7.2. that

$$\lambda\hat{\rho}\frac{\psi_{k-1}'(\lambda\hat{\rho})}{\psi_{k-1}(\lambda\hat{\rho})} - 1 < 0.$$

But because $\psi_{k-1}(\lambda\hat{\rho}) = \hat{\rho}$, the above inequality is equivalent to $\lambda\psi_{k-1}'(\lambda\hat{\rho}) < 1$, as required.

Now we prove the result with respect to k. Since $\hat{\rho}$ is the largest solution to the equation $x = \psi_{k-1}(\lambda x)$ at the point where $\psi_{k-1}(\lambda x)$ crosses x *from below,* it suffices to prove that for any fixed λ

$$\psi_{k-1}(\lambda x) > \psi_k(\lambda x)$$

for all $x > 0$ (if ψ_k shifts down then the largest solution shifts up because the intersection is from below). But this is immediate from the definition of ψ_k, since

$$\psi_{k-1}(\lambda x) - \psi_k(\lambda x) = \mathbb{P}(\mathrm{Po}(\lambda x) \geq k-1) - \mathbb{P}(\mathrm{Po}(\lambda x) \geq k) = \mathbb{P}(\mathrm{Po}(\lambda x) = k-1) > 0,$$

and the result is proved.

A.2 Proof of Theorem 2

It is a well known property of the Poisson degree distribution that site percolation scales the mean of the distribution (e.g. see Campbell et al., 2024, Appendix A). That is, if the degree distribution of the graph is $\mathrm{Po}(\lambda)$, then site percolation with occupation probability σ induces a $\mathrm{Po}((1-\sigma)\lambda)$ degree distribution. Note that because σ is the probability of *removing* a vertex, the rescaled distribution has mean $(1-\sigma)\lambda$ rather than $\sigma\lambda$. It follows from theorem 1 that $\mathrm{Po}((1-\sigma)\lambda)$ has a giant k-core if and only if $(1-\sigma)\lambda > c_k$, that is, if and only if $\sigma < 1 - c_k/\lambda$, as claimed.

A.3 Proof of Corollary 1

Follows immediately from proposition 1 and the fact that the degree is $\mathrm{Po}((1-\sigma)\lambda)$ distributed.